

Module 5

Information Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Social engineering is the art of manipulating people into performing desired actions. It exploits the human desire to be helpful and the natural human instinct to be trusting.

Answer 1(b)

Examples of social engineering include:

- Attempting to trick a user into revealing their password by pretending to be in IT support.
- Gaining access to secure locations by following authorised users through secure doors.
- Leaving USB flash drives or other removable media containing malware lying around in the hope a user will use it and infect a computer.
- Using e-mail, text messages and other means of communication to trick a user into going to a malicious website or installing malware.
- Searching an organisation's dumpsters and trash bins for sensitive information or information that could assist with another type of social engineering attack.

[Any appropriate answer(s) is/ are acceptable.]

[Any three of the above]

Answer 1(c)

- (i) If the current password is compromised, this measure can reduce the risk of a hacker using the compromised password continuously to access sensitive data or system functions. The hacker is thus required to spend extra time and effort to crack the new password.
- (ii) This measure can stop a hacker from guessing a user's password with brute-force attacks, where the hacker uses different combinations of random characters until the password is guessed, or dictionary attacks, where the hacker uses different common passwords or vocabulary words from dictionaries until the password is guessed. The victimised user may even receive a notification message that his/ her account is being locked as someone has tried to log in to his/her account with an incorrect password.
- (iii) This measure can make the passwords harder to guess or crack. A hacker is required to guess a password with more combinations of random characters, and thus more time will be spent in guessing a new correct password.

Answer 2(a)

A company can develop, achieve or maintain competitive advantages in the following areas:

- Cost leadership: Offer products or services at the lowest cost in the industry.
- Differentiation: Offer unique products or services or deliver them in a way that is different and better than what competitors offer.
- Innovation: Introduce new products and services, add new features to existing products and services or invent new ways to produce them.
- Operational effectiveness: Improve the way an organisation executes its internal business processes so that it performs these activities more effectively than its rivals.
- Customer orientation: Concentrate on satisfying customers and establishing customer loyalty.

[Any three of the above]

Answer 2(b)

Management information systems can generate the following types of reports:

- periodic reports;
- exception reports; and
- ad hoc, or on demand reports.

Answer 2(c)

- Periodic reports are created or run according to a pre-set schedule, such as a daily or weekly sales report, or a monthly or quarterly income statement.

[Any appropriate example is acceptable.]

- Exception reports are generated only when something is abnormal or irregular, either higher or lower than the expected or acceptable level, such as an exception report which shows defective parts which exceed the predetermined defect percentage, or an exception report which shows unprofitable sales items or transactions.

[Any appropriate example is acceptable.]

- Ad hoc or on demand reports are unplanned reports generated on demand to learn more about a specific situation or problem, such as an inventory loss report of a robbery or a report of lost items during an earthquake.

[Any appropriate example is acceptable.]

Answer 2(d)

The two differences between predictive data analytics and prescriptive data analytics are as follows:

- Predictive analytics do not predict one possible future but rather multiple future outcomes based on the decision maker's actions; while prescriptive analytics attempt to quantify the effect of future decisions to advise on possible outcomes before the decisions are made.
- Prescriptive analytics go beyond descriptive and predictive models by recommending one or more courses of action and showing the likely outcome of each decision.

Answer 3(a)

Primary activities are related to the production and distribution of the products and services which customers purchase and pay.

Secondary activities, such as HR, administration and management, are those that support the primary activities.

Answer 3(b)

A production department in a manufacturing company has the following primary duties or responsibilities:

- manufacturing products;
- designing, creating and improving production processes;
- preparing production plans and schedules;
- monitoring and improving production quality;
- monitoring and improving production efficiency; and
- monitoring and reducing production costs.

[Any two of the above]

A production department needs the following information to support its duties and responsibilities:

- product designs and specifications;
- inventory records;
- employee schedules;
- labour costs;
- logistics data;
- maintenance records and schedules;
- facility and equipment availability; and/ or
- data collected on the shop floor from process and quality control systems.

[Any two of the above]

Answer 3(c)

A computer network can be classified into the following basic functions:

- Communication – this provides sufficient capacity for human and machine-generated transmissions;
- Search – this is able to locate data, contracts, documents, spreadsheets and other knowledge within an organisation easily and efficiently;
- Mobility – this provides secure, trusted and reliable access from any mobile device anywhere at satisfactory download and upload speeds;
- Relationships – this manages interaction with customers, supply chain partners, shareholders, employees, regulatory agencies and so on; and
- Collaboration – this supports teamwork that may be synchronous or asynchronous; brainstorming; and knowledge and document sharing.

[Any three of the above]

Answer 3(d)

Cloud computing can be implemented in one of the following three deployment models:

- Public cloud: Services are provided "as a service" over the Internet with little or no control over the underlying technology infrastructure; (Remark: Under a public cloud, multiple clients or subscribers use the shared storage space and/ or the same data processing functions provided by the same cloud-based system/ server.);
- Private cloud: All the system activities and functions are deployed and processed over a company intranet or hosted data centre; or
- Hybrid cloud: Activities and tasks are allocated to a mix of private and public clouds.

Answer 3(e)

Cloud storage services offer the following benefits:

- Scalability: Storage capacity is virtually unlimited and relatively inexpensive. Users can store as much data as they want and pay only for the storage used;
- Convenient access: Because data is stored online, anyone with an Internet-enabled device and the necessary login credentials can access the data from anywhere in the world. Users can easily share files without e-mailing attachments; and
- Disaster recovery: Cloud storage is off-site storage, so data is protected against loss from theft, fire and other catastrophes. Cloud storage providers routinely back up all data to another remote storage location for additional protection.

[Any two of the above]

Answer 4(a)

Apart from the waterfall model, other software development models include:

- Prototyping: An iterative development process focusing on user requirements and implementing portions of the proposed system. Prototypes are "screenshots", which eventually are modelled into a portion of the functioning system;
- Rapid application development: An iterative development process using prototypes and automated development tools, such as PowerBuilder and Visual Basic, to speed up the development process; and
- Agile: A software development methodology in which self-organising, cross-functional teams develop software in iterative, incremental cycles. Instead of engaging in considerable up-front planning, teams gradually build functionality into the software.

[Any two of the above]

Answer 4(b)

Journal entry function or financial reporting function should be classified as an essential feature; it is a capability that the accounting software must have. A buyer may not acquire this accounting software if such an essential function is missing.

Multi-user access function or multi-currency function should be classified as a value adding feature. It is a non-essential capability, but it offers significant advantages for an organisation. Some organisations are willing to pay a premium to buy such a function.

Answer 4(c)

The common sources of system change requests are as follows:

- Problem reports in which system error(s) or bug(s) is/ are identified;
- User requests for enhancements;
- Other system development projects which need system integration between two or more systems;
- System infrastructure or technology changes; and
- Senior management requests for better data protection or security controls.

[Any appropriate answer(s) is/ are acceptable.]

[Any three of the above]

Answer 5(a)

Hong Kong is perfectly suited and situated for e-commerce imports and exports due to the following reasons:

- A long history of international trade;
- The expertise of experienced merchandise traders;
- Advanced information and communications technology (ICT);
- Easy access to Mainland China as a source of low-cost merchandise;
- A free port with good air connectivity; and
- No import tax or value added tax (VAT).

[Any four of the above]

Answer 5(b)

- (i) Marketing and advertising must accurately describe the product/service that customers expect; otherwise, customers are likely to be disappointed. A good rule of thumb is to under-promise and over-deliver, so customers are pleasantly surprised when they receive the product with some accessories or an extend warranty period.
- (ii) With an innovative or complex product or service, customers may require extensive training on how to use it and maximise its benefits. When customers know how to use the product properly and can quickly experience the benefits, they are less likely to return it.

Answer 5(c)

The knowledge management system cycle consists of six steps as below:

- Step 1: Create knowledge;
- Step 2: Capture knowledge;
- Step 3: Refine knowledge;
- Step 4: Store knowledge;
- Step 5: Manage knowledge; and
- Step 6: Disseminate knowledge.

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